

HOMEWORK 1

6500-324: DATA MANAGEMENT FOR INFORMATION SYSTEMS (Fall 2022)

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Part 1. Multiple Choice or True/False

1. (1 point) The mapping cardinality between Publisher and Book is usually
A) 1:m
B) m:1
C) 1:1
D) m:m
E) 1:0
2. (1 point) An attribute must be the following except for which one:
A) a column of a table
B) a record
C) a field
D) single-valued
3. (1 point) What a logical data model is used for:
A) For representing data requirements
B) For representing business logics
C) For representing how data is organized in a database
D) For representing how functions are performed
4. (1 point) (check all that apply) What is the difference between an ERD and a relational model
A) They are used in different stages of the database development
B) An ERD tells how records are organized and a relational model shows data requirements
C) A relational model tells how records are organized and an ERD shows data requirements
D) They use different sets of concepts, terminologies, and notations
E) They use unrelated sets of concepts, terminologies, and notations
5. (1 point) What is not correct about a record?
A) it is made of many values observed from one entity
B) it is usually linked to other records
C) it is made of a single value
D) it is also called a tuple
E) It is usually identifiable through some parts of the record
6. (1 point) What is not true about a hierarchical database?
A) records are linked by pointers
B) records are organized as an inverted tree
C) records are not identified by keys
D) it is not a dominant type of databases nowadays
7. (1 point) A relational database gets its name because

- A) records are stacked into relations
B) all records are related
C) records are linked by relationships, rather than pointers
D) Edgar F. (Ted) Codd named it as such for no reason
E) All records are called relations
8. (1 point) (check all that apply) Identify Objects for Data Modeling: Which of the following are objects for the problem domain of inventory management?
A) products, warehouses, inventories, accounts
B) orders, shipments, invoices, packingslips, employees, locations
C) price, quantity, balance, business, services
D) minimum reorder level, balance, inventory system, company
E) suppliers, checkouts, checkins
9. (1 point) (check all that apply) Identify Object Attributes: Which of the following are attributes of orders in an e-commerce system?
A) order date
B) promised date
C) ship date
D) total order amount
E) order quantity
10. (1 point) In the relational model, relationships between relations or tables are created by using:
A) composite keys.
B) determinants.
C) candidate keys.
D) foreign keys.